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Bo He^a, Alain Togbé^b & Shichun Yang^a

^a Department of Mathematics, ABa Teacher's College, Wenchuan, Sichuan, 623000, P.R. China

^b Mathematics Department, Purdue University North Central, 1401 S, U.S. 421, Westville, IN, 46391, USA

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ON THE SOLUTIONS OF THE EXPONENTIAL DIOPHANTINE EQUATION $a^x + b^y = (m^2 + 1)^z$

BO HE

*Department of Mathematics, ABa Teacher's College, Wenchuan, Sichuan, 623000,
P.R. China.*

E-Mail bhe@live.cn

ALAIN TOGBÉ

*Mathematics Department, Purdue University North Central, 1401 S, U.S. 421,
Westville, IN 46391, USA.*

E-Mail atogbe@pnc.edu

SHICHUN YANG

*Department of Mathematics, ABa Teacher's College, Wenchuan, Sichuan, 623000,
P.R. China.*

E-Mail ysc1020@sina.com

ABSTRACT. In this paper, we consider the Diophantine equation $a^x + b^y = c^z$. Combining Laurent's result on lower bounds for linear forms in two logarithms, Bugeaud's result on upper bounds for the p -adic logarithms, and Bilu-Hanrot-Voutier's result on primitive divisors of Lucas numbers, we obtain a sharper computable upper bound for the number of positive integer solutions of the equation $|v_r|^x + |u_r|^y = c^z$, where $v_r^2 + u_r^2 = (m^2 + 1)^r$ and r is an odd number. Moreover, if r is a prime and $r \equiv 5 \pmod{8}$ or $r \equiv 19 \pmod{24}$, $m \geq \frac{2}{\pi}r$, we prove that the equation has only the solution $(x, y, z) = (2, 2, r)$.

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Key words: Exponential Diophantine equation, Terai conjecture, positive integer solution, linear forms in two logarithms, lower bound, primitive divisor.

1. Introduction and the main result. Let $\mathbb{Z}, \mathbb{N}, \mathbb{P}$ be the sets of all integers, positive integers and primes respectively. Let a, b, c be fixed coprime positive integers with $\min(a, b, c) > 1$. The equation

$$a^x + b^y = c^z, \quad x, y, z \in \mathbb{N} \tag{1}$$

is an important and general exponential Diophantine equation which attracts the attention of many authors (see [6]–[27], [29]–[32]). In 1933, Mahler [27] used his p -adic analogue method of the method of Thue-Siegel to prove that equation (1) has only finitely many solutions (x, y, z) . His method is ineffective. An effective result for solutions of equation (1) was given by Gel'fond [15] in 1940. If $x = y = p$, then equation (1) becomes the Catalan-Fermat equation $x^p + y^p = z^q$, see [10], [14].

The case $p = 2$ is an important ingredient to the topic. Essentially this case has been solved in a quite general form by Ellenberg [14].

In 1994, Terai [29] conjectured that if equation (1) has a solution $(x, y, z) = (p, q, r)$, with $\min(p, q, r) > 1$, then equation (1) has only the solution $(x, y, z) = (p, q, r)$. But the conjecture is clearly false (see [6], [7], [21]). On the other hand, heuristics show that equation (1) has at most one solution (x, y, z) , with $\min(x, y, z) > 1$ (see [23], [8]). The conjecture was proved for some special cases. But, in general, the problem is still unsolved.

For $r > 1$, assume that a, b, c satisfying

$$a^2 + b^2 = c^r = (m^2 + n^2)^r, \quad (2)$$

where $2 \mid m$, $\gcd(m, n) = 1$. Then equation (1) has a solution $(x, y, z) = (2, 2, r)$ with $\min(x, y, z) > 1$. For $r = 2$, Jeśmanowicz set a conjecture related to Pythagorean numbers, whose particular cases were discussed by many authors. One can cite for examples [11], [19], [25]. For $r = 3, 5$ and $n = 1$, when b or c are odd prime or prime power, equation (1) was solved by Terai [29], [30], Cao [6], Dong and Cao [13], Le [20], etc. For any b and c , using Gel'fond-Baker method, Le [21] proved that if $r = 5$, $n = 1$, and $m \geq 524$, then equation (1) has only the solution $(x, y, z) = (2, 2, 5)$. When $r = 3, 5$ and $n = 1$, equation (1) was solved by Cao and Dong [7], Hu and Yuan [17] etc. Here is a quick summary of conditions for which equation (1) has only the solution $(x, y, z) = (2, 2, r)$:

- (Cao and Dong [7]): $a \equiv 2 \pmod{4}$, $b \equiv 3 \pmod{4}$, $n = 1$, and $r < \frac{m}{\sqrt{825 \log(m^2+1)}}$, $m > 200$.
- (Le [22]): $a > b$, $a \equiv 3 \pmod{4}$, $b \equiv 2 \pmod{4}$, $n = 1$, and $\frac{a}{b} > (e^{r/1856} - 1)^{-1/2}$.
- (Le [23]): $a \equiv 2 \pmod{4}$, $b \equiv 3 \pmod{4}$, and $c > 3.10^{27}$, $r > 7200$.
- (Le [24]): $r \equiv 5 \pmod{8}$ and $n = 1$, $m > r^2$, $r < 11500$, or $m > \frac{2r}{\pi}$, $r > 11500$.
- (Xia and Yuan [32]): $m \equiv 2 \pmod{4}$, $b \equiv 3 \pmod{4}$, $n = 1$, and $r < \frac{m}{\sqrt{1.5 \log_3(m^2+1)-1}}$.

Recently, Cipu and Mignotte [8] studied equation (1) with a, b, c such that, $r, z > 1$ odd, and $a \equiv 2 \pmod{4}$, $b \equiv 3 \pmod{4}$, $\gcd(a, b) = 1$. They improved some of the above results and others.

In this paper, we consider equation (1) with a, b, c verifying equation (2) and $n = 1$. Let

$$v_r + u_r \sqrt{-1} = (m + \sqrt{-1})^r, \quad 2 \nmid r, \quad 2 \mid m, \quad (3)$$

and $a = |v_r|$, $b = |u_r|$, $c = m^2 + 1$, then equation (1) becomes

$$|v_r|^x + |u_r|^y = (m^2 + 1)^z. \quad (4)$$

Using the Gel'fond-Baker method, Le [26] removed the restriction that $a \equiv 2 \pmod{4}$ and $b \equiv 3 \pmod{4}$ and proved that if $m > 10^6 r^6$, then equation (4) has only the positive integer solution $(x, y, z) = (2, 2, r)$, which we improve in this paper. Moreover, our method is slightly different of his method. In fact, combining Laurent's result (see [18]) on lower bounds for linear forms in two logarithms and Bugeaud's result (see [4]) on upper bounds of linear forms in the p -adic logarithms, we get a sharper computable upper bound of the number of solutions to equation (4). We prove the following results.

THEOREM 1.1. *If $m > 24650 r^2 (\log r)^2$, then equation (4) has only the positive integer solution $(x, y, z) = (2, 2, r)$.*

When $r \equiv 5 \pmod{8}$ or $r \equiv 3, 19 \pmod{24}$, following ideas of Le [24], Cao [7] and using Bilu-Hanrot-Voutier's result ([1], [3], [31]) on primitive divisors of Lucas numbers, we find sharper bounds in some particular cases. We have

THEOREM 1.2. *If $r \equiv 5 \pmod{8}$ or $r \equiv 3, 19 \pmod{24}$ and $m \geq 2r$, then equation (4) has only the solution $(x, y, z) = (2, 2, r)$.*

THEOREM 1.3. *If r is a prime, $r \equiv 5 \pmod{8}$ or $r \equiv 19 \pmod{24}$, and $m \geq \frac{2}{\pi} r$, then equation (4) has only the solution $(x, y, z) = (2, 2, r)$.*

The organization of this paper is as follows. In Section 2, we recall some useful results particularly a recent result of Laurent [18] on lower bounds of linear forms in two logarithms and a result of Bugeaud [4] on upper bounds of linear forms in p -adic logarithms. Moreover, we prove that if r is an odd positive integer and p is a prime divisor of $b = u_r$, then $p \geq 11$. The proof of Theorem 1.1 will be done in Section 3 by the means of the results of Laurent and Bugeaud cited above. We use the same method to prove Theorems 1.2 and 1.3 in the last section.

2. Preliminary work. In this section, we will recall and prove some results that we will use later to prove our main theorems cited in Section 1. We know that there are many results in the literature to obtain a lower bound for a linear form in logarithms. But we choose the following result due to Laurent (see [18], Theorem 2) on linear forms in two logarithms because it gives us the best estimate. So for any non-zero algebraic number α of degree d over $\mathbb{Q}$, whose minimal polynomial over $\mathbb{Z}$ is $a \prod_{j=1}^d (X - \alpha^{(j)})$, we denote by

$$h(\alpha) = \frac{1}{d} \left(\log |a| + \sum_{j=1}^d \log \max \left(1, |\alpha^{(j)}| \right) \right)$$

the absolute logarithmic height of α . In the case $d = 1$, the logarithm height of the algebraic number is the logarithm of the maximum between the absolute value of the numerator and the absolute value of the denominator. That is what we will use in the remaining sections. Here we integrally recall Laurent's result.

LEMMA 2.1. (Theorem L) *Let a_1, a_2, h, ϱ , and μ be real numbers with $\varrho > 1$ and $\frac{1}{3} \leq \mu \leq 1$. Set*

$$\sigma = \frac{1 + 2\mu - \mu^2}{2}, \quad \lambda = \sigma \log \varrho, \quad H = \frac{h}{\lambda} + \frac{1}{\sigma},$$
$$\omega = 2 \left(1 + \sqrt{1 + \frac{1}{4H^2}} \right), \quad \theta = \sqrt{1 + \frac{1}{4H^2}} + \frac{1}{2H}.$$

Consider the linear form

$$\Lambda_1 = b_2 \log \alpha_2 - b_1 \log \alpha_1,$$

where b_1 and b_2 are positive integers. Suppose that α_1, α_2 are multiplicatively independent. Put

$$D = [\mathbb{Q}(\alpha_1, \alpha_2) : \mathbb{Q}] / [\mathbb{R}(\alpha_1, \alpha_2) : \mathbb{R}],$$

and assume that

$$h \geq \max \left\{ D \left(\log \left(\frac{b_1}{a_2} + \frac{b_2}{a_1} \right) + \log \lambda + 1.75 \right) + 0.06, \lambda, \frac{D \log 2}{2} \right\},$$
$$a_i \geq \max \{ 1, (\varrho - 1) \log |\alpha_i| + 2Dh(\alpha_i) \} \quad (i = 1, 2),$$

and

$$a_1 a_2 \geq \lambda^2.$$

Then we have the lower bound

$$\begin{aligned} \log |\Lambda_1| \geq & -C \left(h + \frac{\lambda}{\sigma} \right)^2 a_1 a_2 - \sqrt{\omega \theta} \left(h + \frac{\lambda}{\sigma} \right) \\ & - \log \left(C' \left(h + \frac{\lambda}{\sigma} \right)^2 a_1 a_2 \right), \end{aligned} \tag{5}$$

with

$$C = \frac{\mu}{\lambda^3 \sigma} \left(\frac{\omega}{6} + \frac{1}{2} \sqrt{\frac{\omega^2}{9} + \frac{8\lambda\omega^{5/4}\theta^{1/4}}{3\sqrt{a_1 a_2} H^{1/2}}} + \frac{4}{3} \left(\frac{1}{a_1} + \frac{1}{a_2} \right) \frac{\lambda \omega}{H} \right)^2, \quad C' = \sqrt{\frac{C \sigma \omega \theta}{\lambda^3 \mu}}.$$

Let p be a prime and v_p denote the standard p -adic valuation normalized by $v_p(p) = 1$. Let $\frac{x_1}{y_1}, \frac{x_2}{y_2}$ be two non-zero rational numbers and let g denote the smallest positive integer such that

$$v_p \left(\left(\frac{x_1}{y_1} \right)^g - 1 \right) > 0, \quad v_p \left(\left(\frac{x_2}{y_2} \right)^g - 1 \right) > 0.$$

If $p = 2$ and

$$v_2 \left(\frac{x_2}{y_2} - 1 \right) \geq 2.$$

Assume that there exists a real number E such that $v_p((\frac{x_1}{y_1})^g - 1) \geq E > \frac{1}{p-1}$. Bugeaud [4] obtained explicit upper bounds for the p -adic valuation of

$$\Lambda_2 = \left(\frac{x_1}{y_1}\right)^{b_1} - \left(\frac{x_2}{y_2}\right)^{b_2},$$

where b_1 and b_2 are positive integers. We have

LEMMA 2.2. (Theorem B) *Let $A_1 > 1, A_2 > 1$ be real numbers such that*

$$\log A_i \geq \max\{\log |x_i|, \log |y_i|, E \log p\}, \quad i = 1, 2, \quad (6)$$

and we put

$$b' = \frac{b_1}{\log A_2} + \frac{b_2}{\log A_1}.$$

If $\frac{x_1}{y_1}$ and $\frac{x_2}{y_2}$ are multiplicatively independent then we have the upper estimates

$$v_p(\Lambda_2) \leq \frac{36.1g}{E^3(\log p)^4} \left(\max \left\{ \log b' + \log(E \log p) + 0.4, 6E \log p, 5 \right\} \right)^2 \log A_1 \log A_2. \quad (7)$$

Proof. See Theorem 2 of [4]. $\square$

LEMMA 2.3. *Let (x, y, z) be a solution of (4). If $r \equiv 3 \pmod{8}$, $m > \frac{2r}{\pi}$, then $2 \mid x$ and $2 \mid y$.*

Proof. From Theorem 1 of [7], we suppose that $r \geq 11$. Let $\varepsilon = m + \sqrt{-1}$, $\bar{\varepsilon} = m - \sqrt{-1}$, and $\theta = \arctan \frac{1}{m}$. Then from equation (3) we have $\varepsilon = \sqrt{c}(\cos \theta + \sqrt{-1} \sin \theta)$. So $a = v_r = c^{\frac{r}{2}} \cos(r\theta)$, $b = u_r = c^{\frac{r}{2}} \sin(r\theta)$. As $m > \frac{2r}{\pi}$, we have $v_r > 0$ and $u_r > 0$.

If $x = 1$, then taking equation (4) mod m^2 , we get $-rm \pm 1 \equiv 1 \pmod{m^2}$; so $m \mid r$. As $m > \frac{2r}{\pi}$, we have $m = r$, which is impossible, so $x \geq 2$. We take again equation (4) mod 4 to obtain $(-1)^y \equiv 1 \pmod{4}$. This means that $2 \mid y$.

Put $E_r = \frac{\varepsilon^r + \bar{\varepsilon}^r}{\varepsilon + \bar{\varepsilon}}$, $F_r = \frac{\varepsilon^r - \bar{\varepsilon}^r}{\varepsilon - \bar{\varepsilon}}$. Then we have $v_r = mE_r$, $u_r = F_r$. Suppose $r = 8k + 3$, where $k \in \mathbb{N}$. Thus we get

$$E_r = E_{8k+3} = \sum_{j=0}^{4k+1} (-1)^j \binom{8k+3}{2j} m^{8k+2-2j} \equiv \begin{cases} 1 \pmod{8}, & \text{if } 2 \mid m, \\ 5 \pmod{8}, & \text{if } 4 \mid m, \end{cases} \quad (8)$$

$$F_r = F_{8k+3} = \sum_{j=0}^{4k+1} (-1)^j \binom{8k+3}{2j+1} m^{8k+2-2j} \equiv \begin{cases} 3 \pmod{8}, & \text{if } 2 \mid m, \\ 7 \pmod{8}, & \text{if } 4 \mid m, \end{cases} \quad (9)$$

and

$$E_r + F_r = E_{8k+3} + F_{8k+3} = \frac{2(\varepsilon^{8k+4} - \bar{\varepsilon}^{8k+4})}{\varepsilon^2 - \bar{\varepsilon}^2} = 4AB, \quad (10)$$

where $A = \frac{\varepsilon^{4k+2} + \bar{\varepsilon}^{4k+2}}{2}$, $B = \frac{\varepsilon^{4k+2} - \bar{\varepsilon}^{4k+2}}{\varepsilon^2 - \bar{\varepsilon}^2}$. It is clear that $A, B \in \mathbb{N}$. Since

$$\begin{aligned} A &= \sum_{j=0}^{2k+1} (-1)^j \binom{4k+2}{2j} m^{4k+2-2j} \\ &\equiv - \sum_{j=0}^{2k+1} \binom{4k+2}{2j} = -2^{4k+1} \pmod{m^2+1}, \end{aligned} \quad (11)$$

$$\begin{aligned} B &= \frac{1}{2} \sum_{j=0}^{2k} (-1)^j \binom{4k+2}{2j+1} m^{4k-2j} \\ &\equiv \frac{1}{2} \sum_{j=0}^{2k} \binom{4k+2}{2j+1} = 2^{4k} \pmod{m^2+1}, \end{aligned} \quad (12)$$

and $A \equiv 3 \pmod{4}$, $B \equiv 1 \pmod{4}$, then from (8), (9), (10), (11) and (12), we get

$$\begin{aligned} \left(\frac{E_r}{F_r}\right) &= \left(\frac{F_r}{E_r}\right) = \left(\frac{E_r + F_r}{E_r}\right) = \left(\frac{AB}{E_r}\right) = \left(\frac{E_r}{A}\right) \left(\frac{E_r}{B}\right) \\ &= \left(\frac{2A \cdot \frac{\varepsilon^{4k+1} + \bar{\varepsilon}^{4k+1}}{\varepsilon + \bar{\varepsilon}} - (m^2+1)^{4k+1}}{A}\right) \left(\frac{B(\varepsilon^{4k+1} - \bar{\varepsilon}^{4k+1}) + (m^2+1)^{4k+1}}{B}\right) \\ &= \left(\frac{-1}{A}\right) \left(\frac{m^2+1}{A}\right) \left(\frac{m^2+1}{B}\right) = - \left(\frac{A}{m^2+1}\right) \left(\frac{B}{m^2+1}\right) \\ &= - \left(\frac{2}{m^2+1}\right) = \begin{cases} 1, & \text{if } 2 \nmid m, \\ -1, & \text{if } 4 \mid m. \end{cases} \end{aligned} \quad (13)$$

We suppose that $m = 2^s m_1$, where $2 \nmid m_1$. As r is odd, we obtain

$$\begin{aligned} \left(\frac{m}{F_r}\right) &= \left(\frac{2^s m_1}{F_r}\right) = \left(\frac{2^s}{F_r}\right) \left(\frac{m_1}{F_r}\right) = \left(\frac{2^s}{F_r}\right) \left(\frac{F_r}{m_1}\right) (-1)^{\frac{m_1-1}{2}} \\ &= \left(\frac{2^s}{F_r}\right) \left(\frac{-1}{m_1}\right) (-1)^{\frac{m_1-1}{2}} = \left(\frac{2^s}{F_r}\right) = \begin{cases} -1, & \text{if } 2 \nmid m, \\ 1, & \text{if } 4 \mid m. \end{cases} \end{aligned} \quad (14)$$

Then

$$\left(\frac{v_r}{u_r}\right) = \left(\frac{mE_r}{F_r}\right) = \left(\frac{m}{F_r}\right) \left(\frac{E_r}{F_r}\right) = -1. \quad (15)$$

One can see that $\left(\frac{c}{u_r}\right) = \left(\frac{u_r}{m^2+1}\right) = \left(\frac{F_r}{m^2+1}\right) = \left(\frac{-2^{r-1}}{m^2+1}\right) = 1$. Therefore, from (15) and (4), we obtain that $2 \nmid x$. This completes the proof of Lemma 2.3. $\square$

LEMMA 2.4. Let $n \in \mathbb{Z}_{>0}$ with $n \leq 10^7$. If $n \notin \{1, 2, 4\}$, then equation

$$X^2 + Y^{2n} = Z^3$$

has no solutions.

Proof. See Corollary 2 of [9]. □

LEMMA 2.5. If $n \in \mathbb{N}$ with $n \geq 4$, then the equation

$$X^n + Y^n = 3Z^2, \quad X, Y, Z \in \mathbb{Z}$$

has no solutions in nonzero pairwise coprime integers (X, Y, Z) with $X > Y$.

Proof. See Theorem 1.1 of [2]. □

LEMMA 2.6. Suppose that $k \in \mathbb{N}$ with $k > 1$. Then the equations

$$X^{2k} + Y^2 = Z^4, \quad X, Y, Z \in \mathbb{Z}, \quad 2 \mid X, \quad \gcd(X, Y) = 1,$$

and

$$X^{2k} + Y^4 = Z^2, \quad X, Y, Z \in \mathbb{Z}, \quad \gcd(X, Y) = 1,$$

have no solutions with $XY \neq 0$.

Proof. See Lemma 9 and Lemma 10 of [7]. □

LEMMA 2.7. Let p be a prime with $p \equiv 3 \pmod{8}$. Then the equation

$$X^4 \pm Y^4 = pZ^2, \quad X, Y, Z \in \mathbb{Z}, \quad \gcd(X, Y) = 1,$$

has no solution with $Z \neq 0$.

Proof. From the result on page 33 in [5], we can obtain that the equation $X^4 - Y^4 = pZ^2$ has no solution with $Z \neq 0$. Since $p \equiv 3 \pmod{4}$, the above equation is impossible because such a prime cannot divide a sum of two coprime squares. Thus, the equation $X^4 + Y^4 = pZ^2$ has no solution with $Z \neq 0$. □

LEMMA 2.8. Let $n > 1$ be an odd integer. Then every solution (X, Y, Z) of the equation

$$X^2 + Y^2 = Z^n, \quad X, Y, Z \in \mathbb{N}, \quad \gcd(X, Y) = 1, \quad 2 \mid X, \tag{16}$$

can be expressed as

$$X + Y\sqrt{-1} = \lambda_1(X_1 + \lambda_2 Y_1 \sqrt{-1})^n, \quad Z = X_1^2 + Y_1^2,$$

where $\gcd(X, Y) = 1$, $\gcd(X_1, Y_1) = 1$, $2 \mid X_1$, $\lambda_1, \lambda_2 \in \{1, -1\}$.

Proof. See pages 122–123 of [28]. $\square$

Now we prove the following result.

LEMMA 2.9. *If r is odd, and p is a prime divisor of $b = u_r$. Then $p \geq 11$.*

Proof. Let u_r be the r th member of a Lucas sequence of roots $\varepsilon = m + \sqrt{-1}$ and $\bar{\varepsilon} = m - \sqrt{-1}$. So its discriminant is $\Delta = (\varepsilon - \bar{\varepsilon})^2 = -4$. Since r is an odd prime, any prime factor p of u_r either divides Δ (which is false because u_r is odd), or it is primitive for u_r , so $p \equiv \left(\frac{\Delta}{p}\right) \pmod{2r}$. So, we see that $p < 11$ can happen only if $r = 3$ and $p = 5 \equiv -1 \pmod{3}$, which cannot happen because $\left(\frac{-4}{5}\right) = -1$, or $p = 7 \equiv 1 \pmod{3}$, which cannot happen either because $\left(\frac{-4}{7}\right) = 1$. Therefore, $p \geq 11$. $\square$

LEMMA 2.10. *Suppose $r \equiv 5 \pmod{8}$ or $r \equiv 3, 19 \pmod{24}$, $m > \frac{2r}{\pi}$. Let (x, y, z) be a solution of equation (4) with $(x, y, z) \neq (2, 2, r)$. Then $x = 2$, $2 \mid y$, $y \geq 6$, and $2 \nmid z$.*

Proof. When $r \equiv 5 \pmod{8}$, see formula (54) in the proof of Theorem in [24].

So suppose $r \equiv 3, 19 \pmod{24}$. Using Lemma 2.3, one can see if $m > \frac{2r}{\pi}$ and if (x, y, z) is a solution of equation (3), then $2 \mid x$ and $2 \mid y$.

Assume that $x \geq 4$. Taking equation (4) mod m^4 , we get $\frac{1}{2}r(r-1)y + z \equiv 0 \pmod{m^2}$. So $2 \mid z$. Put $x = 2x_1$, $y = 2y_1$, and $z = 2z_1$. Clearly, if $2 \mid y_1$, then $2 \mid z_1$, or if $2 \mid z_1$, then $2 \mid y_1$. This contradicts Lemma 2.6. Therefore, $2 \nmid y_1 z_1$. Equation (4) implies

$$v_r^{2x_1} = (c^{z_1} + u_r^{y_1})(c^{z_1} - u_r^{y_1}). \quad (17)$$

Since

$$c^{z_1} + u_r^{y_1} \equiv 0 \pmod{m^2}, \quad c^{z_1} - u_r^{y_1} \equiv 2 \pmod{m^2}, \quad (18)$$

from equation (17) we deduce

$$c^{z_1} + u_r^{y_1} = \frac{1}{2}m^{2x_1}s^{2x_1}, \quad c^{z_1} - u_r^{y_1} = 2t^{2x_1}, \quad (19)$$

where $s, t \in \mathbb{N}$, $st = \frac{v_r}{m}$, and $\gcd(s, t) = 1$. Hence, if $3 \nmid m$, then $m^2 \equiv 1 \pmod{3}$. By equation (4) and Lemma 2.9, we have

$$v_r^{2x_1} \equiv \left(\sum_{k=0}^{(r-1)/2} (-1)^k \binom{r}{2k+1} \right)^{2x_1} \equiv 1 \pmod{3}, \quad u_r^{2x_1} \equiv 1 \pmod{3},$$

and $c^{2z_1} \equiv 1 \pmod{3}$. This is a contradiction. Therefore, $3 \mid m$. Since $r \equiv 3, 19 \pmod{24}$ and $\frac{1}{2}r(r-1)y + z \equiv 0 \pmod{m^2}$, we get $3 \mid z$. Then from (19) we obtain

$$t^{2x_1} + \frac{1}{4}m^{2x_1}s^{2x_1} = (c^{z_1/3})^3. \quad (20)$$

Also Lemma 2.8 and equation (20) imply

$$\frac{1}{2}m^{x_1}s^{x_1} = X_1|X_1^2 - 3Y_1^2|, \quad t^{x_1} = Y_1|Y_1^2 - 3X_1^2|, \quad (21)$$

where $X_1, Y_1 \in \mathbb{N}$, $\gcd(X_1, Y_1) = 1$, and $2|X_1, 2 \nmid Y_1$. Since $3|m$, from (21) we see that $3|X_1$. Hence, $\gcd(Y_1, Y_1^2 - 3X_1^2) = \gcd(Y_1, 3X_1^2) = 1$. Thus, using again (21), we have $Y_1 = U_1^{x_1}, |Y_1^2 - 3X_1^2| = U_2^{x_1}$, and so

$$U_1^{2x_1} \pm U_2^{x_1} = 3X_1^2, \quad (22)$$

where $U_1, U_2 \in \mathbb{N}$, $\gcd(U_1, U_2) = 1$, $U_1U_2 = t$, and $2 \nmid U_1 + U_2$.

Since $x \geq 4$, we have $p|x_1$, where p is a prime. If $p = 3$, then $6|x$ and $6|z$. Lemma 2.4 implies that equation (4) has no solutions. If $p \geq 5$, then (22) is impossible by Lemma 2.5. When $x_1 = 2^l$, where $l \in \mathbb{N}$, $l \geq 1$, then equation (4) becomes $v_r^{2^{l+1}} + u_r^{2y_1} = (m^2 + 1)^{2z_1}$. If $l = 1$, then $y_1 = 1$ (see Lemma 2.6). From equations $v_r^4 + u_r^2 = (m^2 + 1)^z$ and $v_r^2 + u_r^2 = (m^2 + 1)^r$, we see that $(m^2 + 1)^r | v_r^2 - 1$, which is a contradiction. If $l \geq 2$, then equation (22) has also no solutions by Lemma 2.7. Hence $x = 2$.

Because $2|y$, and $(x, y, z) \neq (2, 2, r)$, then $y \geq 4$. Again using equations $v_r^2 + u_r^4 = (m^2 + 1)^z$ and $v_r^2 + u_r^2 = (m^2 + 1)^r$, we see that $(m^2 + 1)^r | v_r^2 - 1$. Thus $y \geq 6$. Taking equation (4) mod m^4 , we get $r^2 \equiv \frac{1}{2}r(r-1)y + z \pmod{m^2}$. Therefore, $2 \nmid z$. This completes the proof of Lemma 2.10. $\square$

LEMMA 2.11. *Let (X, Y, Z) be a solution of equation (16). If $2 \nmid n$, $n > 18460$, and $Z > 1.37 \times 10^8$, then $\min(X, Y) > Z^{n/3}$.*

Proof. See Lemma 10 and its proof in [24]. The conclusion comes from formula (37) of [24]. $\square$

REMARK 2.12. In Lemma 10 of [24], the lower bound of n found by Le is 11500. This is wrong. In fact, the lower bound of n should be 18460. Therefore, the conclusion of this paper should be accordingly modified.

A Lucas pair (α, β) is a pair of algebraic integers α, β such as that $\alpha + \beta$ and $\alpha\beta$ are nonzero coprime rational integers, and $\frac{\alpha}{\beta}$ is not a root of unity. For a given Lucas pair (α, β) , one defines the corresponding sequence of *Lucas numbers* by

$$u_n(\alpha, \beta) = \frac{\alpha^n - \beta^n}{\alpha - \beta}, \quad 2 \nmid n, \quad n = 1, 3, \dots$$

DEFINITION 2.13. Let p be prime. The prime p is a *primitive divisor* of a Lucas number $u_n(\alpha, \beta)$, if $p | u_n(\alpha, \beta)$ and $p \nmid (\alpha - \beta)^2 u_1(\alpha, \beta) \cdots u_{n-1}(\alpha, \beta)$.

LEMMA 2.14. *For any prime p not dividing $\alpha\beta$, there exists a positive integer m such that $p | u_m(\alpha, \beta)$. Let m_p be the smallest m with this property. If $p \nmid (\alpha + \beta)^2$, and $p \nmid (\alpha - \beta)^2$, then $p | u_m(\alpha, \beta)$ if and only if $m_p | m$, and $m_p | (p-1)$ or $m_p | (p+1)$.*

Proof. See Corollary 2.2 of [3]. $\square$

LEMMA 2.15. (Theorem BHV) *For any integer $n > 30$, every n th term of any Lucas sequence has a primitive divisor. Further, if $1 < n \leq 30$, $2 \nmid n$, and the Lucas sequence whose n th term has no primitive divisor, then $n = 5, 7, 13$.*

Proof. See Theorem 1.4 of [3] and Theorem 1 of [31]. One can see also [1]. $\square$

3. Proof of Theorem 1.1. We suppose that $m > 24650r^2(\log r)^2$, and from the results of [7] and [17], we suppose that $r \geq 7$. Moreover, from equation (3), we put $\theta = \arctan \frac{1}{m}$, so

$$a = v_r = c^{\frac{r}{2}} \cos(r\theta), \quad b = u_r = c^{\frac{r}{2}} \sin(r\theta). \quad (23)$$

3.1. The upper bound of x, y, z . In this subsection, we will determine some upper bounds for x, y, z using two cases.

3.1.1. Case $a^x \leq b^{\frac{y}{2}}$. If $a^x \leq b^{\frac{y}{2}}$, then from (1), we get

$$z \log c = \log(a^x + b^y) = y \log b + \log \left(1 + \frac{a^x}{b^y} \right) < y \log b + \frac{a^x}{b^y}. \quad (24)$$

Let

$$\Lambda_1 = z \log c - y \log b.$$

We have $\Lambda_1 > 0$ and

$$\log \Lambda_1 < x \log a - y \log b \leq -\frac{1}{2}y \log b. \quad (25)$$

To apply Lemma 2.1, we take

$$D = 1, \quad \alpha_1 = c, \quad \alpha_2 = b, \quad b_1 = z, \quad b_2 = y,$$

and

$$a_1 = (\varrho + 1) \log c, \quad a_2 = (\varrho + 1) \log b, \quad \mu = 0.51, \quad \varrho = 5.1.$$

So we find $\sigma = 0.87995$, $\lambda = 1.4337$. As $r \geq 7$, $m \geq 5 \cdot 10^5$, and using (23), we get $a_1 > 12.2 \log m > 157$. Also $0 < \alpha < \frac{1}{100}$ implies $\sin \alpha > \frac{1}{2}\alpha$. We deduce that $a_2 = 6.1 \log b > 6.1 \log(\frac{m^6}{14}) > 456$. So $a_1 a_2 \geq \lambda^2$. From (24) and the assumption $a^x < b^{y/2}$, we have

$$\frac{z}{\log b} + \frac{y}{\log c} < \frac{2y}{\log c} + 1. \quad (26)$$

Therefore, we obtain

$$\begin{aligned} \max \left\{ \left(\log \left(\frac{b_1}{a_2} + \frac{b_2}{a_1} \right) + \log \lambda + 1.75 \right) + 0.06, \lambda, \frac{\log 2}{2} \right\} \\ = \log \left(\frac{b_1}{a_2} + \frac{b_2}{a_1} \right) + 2.171. \end{aligned}$$

So we consider

$$h = \max \left\{ 8.7, \log \left(\frac{2y}{\log c} + 1 \right) + 0.362 \right\}.$$

If $\log \left(\frac{2y}{\log c} + 1 \right) + 0.362 \leq 8.7$, then $\frac{y}{\log c} < 2090$. If $\log \left(\frac{2y}{\log c} + 1 \right) + 0.362 > 8.7$, then a straightforward computation gives $C = 0.3556$, $C' = 0.9455$ (we just use lower bounds of a_1, a_2, h). Hence inequality (5) becomes

$$\log |\Lambda_1| \geq -13.25(h + 1.63)^2 \log b \log c. \quad (27)$$

Therefore, inequalities (25) and (27) imply

$$\frac{y}{2 \log c} < 13.25 \left(\log \left(\frac{2y}{\log c} + 1 \right) + 1.992 \right)^2. \quad (28)$$

Thus we have $\frac{y}{\log c} < 3040$, i.e.

$$y < 3040 \log c. \quad (29)$$

3.1.2. Case $a^x > b^{\frac{y}{2}}$. If $a^x > b^{\frac{y}{2}}$, then one use $m \geq 24650r^2$, $\tan \alpha > \frac{1}{2}\alpha$, $\sin \alpha > \frac{1}{2}\alpha$ (as $0 < \alpha < \frac{1}{100}$), $r \geq 7$, and (23) to obtain

$$y < \frac{2 \log a}{\log b} x < \frac{2 \log c^{r/2}}{\log(c^{r/2} \sin \frac{r}{2m})} x < \frac{2 \log c^{r/2}}{\log \frac{c^{r/2}}{m}} x < \frac{2r}{r-1} x \leq \frac{7}{3} x. \quad (30)$$

Now using equation (4), we get $c^z = a^x + b^y < a^x + a^{2x} < 1.01a^{2x}$. Because z, r, x are positive integer, thus, we have

$$z < \frac{2x \log a + \log 1.01}{\log c} \leq rx. \quad (31)$$

We suppose s is an integer, p is a prime such that $p^s \mid m$. In Lemma 2.2, we take

$$\Lambda_2 = v_r^x = a^x = \left(\frac{x_1}{y_1} \right)^{b_1} - \left(\frac{x_2}{y_2} \right)^{b_2},$$

where

$$x_1 = c = m^2 + 1, y_1 = 1, x_2 = b = u_r, y_2 = 1, b_1 = z, b_2 = y.$$

Moreover, from (3) we have

$$\gcd(c, b) = \gcd \left(m^2 + 1, \sum_{j=0}^{\frac{r-1}{2}} \binom{r}{2j+1} \right) = \gcd(m^2 + 1, 2^{r-1}) = 1,$$

then $\frac{x_1}{y_1}$ and $\frac{x_2}{y_2}$ are multiplicatively independent. So one can see that

$$v_p(\Lambda_2) \geq sx. \quad (32)$$

Since $v_p((\frac{x_1}{y_1})^2 - 1) > 0$ and $v_p((\frac{x_2}{y_2})^2 - 1) > 0$, we take $g = 2$, $E = 1$, $\log A_1 = \log c$, and $\log A_2 = \log b$. As $c^z > b^y$, we have

$$\frac{z}{\log b} + \frac{y}{\log c} < \frac{2z}{\log b} = b'. \quad (33)$$

We will consider two cases. First, if $m = 2^s$ and as $m \geq 8$, then $v_2(\frac{x_2}{y_2} - 1) \geq 2$. Using (7) and (32), we get

$$sx \leq \frac{72.2}{(\log p)^4} \left(\max \left\{ \log b' + \log(\log p) + 0.4, 6 \log p, 5 \right\} \right)^2 \log b \log c. \quad (34)$$

Therefore, if $m = 2^s$, we take $p = 2$ and $6 \log p < 5$. So the definition of Λ_2 , (33), and (34), we deduce from

$$sx \leq 312.8 \left(\max \left\{ \log \left(\frac{2z}{\log b} \right) + 0.4, 5 \right\} \right)^2 \log b \log c. \quad (35)$$

Second, if $m \neq 2^s$, where s is a positive integer, we take $p \geq 3$, then $6 \log p > 5$. So inequalities (33) and (34) imply

$$sx \leq \frac{72.2}{(\log p)^2} \left(\max \left\{ \frac{1}{\log p} \log \left(\frac{2z}{\log b} \right) + 0.4, 6 \right\} \right)^2 \log b \log c. \quad (36)$$

3.2. The upper bound for m . In this subsection, we will determine an upper bound for m . For that, we distinguish four cases.

3.2.1. Case $x = 1$. As $m \geq 100$, we take equation (4) modulo m^2 and we get $r \equiv 0 \pmod{m}$. Then $m \leq r$, which is a contradiction.

3.2.2. Case $x = 2$. If $y = 1$, then from equation (4) and $a^2 + b^2 = c^r$ we get $b^2 - b = c^z(c^{r-z} - 1)$. So $c^z \mid b - 1$, which is impossible.

If $y = 2$, then equation (4) has the solution $(x, y, z) = (2, 2, r)$.

If $y = 3, 4$, then from equation (4) and $a^2 + b^2 = c^r$ we get $b^2(b^{y-2} - 1) = c^r(c^{z-r} - 1)$. So $c^r \mid b^{y-2} - 1$, which is also impossible. Thus $y \geq 5$.

We know that $\sin \alpha > \frac{1}{2}\alpha$ when $0 < \alpha < \frac{1}{100}$. So from (23) and as $r \geq 7$, we have $b^y \geq b^5 = c^{2.5r} \sin^{2.5}(r\theta) > c^{2.5r} (\frac{r}{2m})^{2.5} > c^{2r} > a^4$. Then one deduces that $c^z = a^2 + b^y < 2b^y$. So we use again (23) to get $c^z = a^2 + b^y < c^r + c^{ry/2} \sin^y(r\theta) < c^r + \frac{1}{2}c^{ry/2} < c^{ry/2}$. Therefore, we obtain $z < \frac{1}{2}ry$. Now we take equation (4) modulo m^4 . If $r \equiv 1 \pmod{4}$, we see that $r^2 - \frac{1}{2}r(r-1)y \equiv z \pmod{m^2}$. If $r \equiv 3 \pmod{4}$, we get $2 \mid y$. We also obtain $r^2 - \frac{1}{2}r(r-1)y \equiv z \pmod{m^2}$. This implies that $\frac{1}{2}r(r-1)y + z \geq m^2 + r^2$. As $\frac{1}{2}r(r-1)y + z < \frac{1}{2}r^2y$ and using (29), we get

$$\frac{2m^2}{r^2} + 2 \leq y \leq 3040 \log(m^2 + 1). \quad (37)$$

We obtain $m < 200r \log r$. This is a contradiction.

3.2.3. Case $x = 3$. If $y = 1, 2$, then from equation (4) and $a^2 + b^2 = c^r$ we get $a^2(ab^{2-y} - 1) = c^r(b^{2-y}c^{r-z} - 1)$ or $c^z(b^{2-y} - c^{z-r})$. Thus $c^r \mid ab^{2-y} - 1$ or $c^z \mid ab^{2-y} - 1$. This is impossible.

If $y \geq 3$, then from equation (4) and $a^2 + b^2 = c^r$, we get $a^2(b^{y-2} - a) = c^r(b^{y-2} - c^{z-r})$. So $c^r \mid b^{y-2} - a$. Therefore, this is impossible when $y = 3, 4$.

If $y = 5$, then we have $c^r = a^2 + b^2 \mid b^3 - a$. We deduce that $a^2 + b^2 \mid ab^3 + b^2$, i.e. $a^2 + b^2 \mid ab + 1$, which is also impossible.

If $y = 6$, then taking equation (4) modulo m^3 , we get $\frac{1}{2}r(r-1)y + z \equiv 0 \pmod{m}$. Thus $2 \mid z$. If $3 \mid m$, then $3 \mid z$. So by the result on Fermat equation $X^3 + Y^3 = Z^3$, equation (4) has no positive integer solution. If $3 \nmid m$, then by Lemma 2.9 and equation (4), we see that $b^6 \equiv 1 \pmod{3}$ and $c^z \equiv 1 \pmod{3}$. Thus $3 \mid a$ and $3 \mid \frac{a}{m}$. From equation (3) and as $m^2 \equiv 1 \pmod{3}$, we get

$$\frac{a}{m} = \sum_{k=0}^{(r-1)/2} (-1)^k \binom{r}{2k} m^{r-2k-1} \equiv \pm 2^{\frac{r-1}{2}} \not\equiv 0 \pmod{3},$$

which is a contradiction. Therefore, $y \geq 7$.

Again here, we use equation (23), the fact that $\sin \alpha > \frac{1}{2}\alpha$ (as $0 < \alpha < \frac{1}{100}$) and $r \geq 7$ to obtain

$$b^y \geq b^7 = c^{3.5r} \sin^{3.5}(r\theta) > c^{3.5r} \left(\frac{r}{2m}\right)^{3.5} > c^{3r} > a^6.$$

Then $c^z = a^3 + b^y < 2b^y$. Similarly, we get $z < \frac{1}{2}ry$. Since $m \geq 100$ and considerations modulo m^3 of equation (4) give $\frac{1}{2}r(r-1)y + z \equiv 0 \pmod{m}$. So $y > \frac{1}{r^2}m$. Therefore, from (29), we get

$$m \leq 3040r^2 \log(m^2 + 1). \quad (38)$$

This implies that $m < 24650r^2(\log r)^2$, which is also a contradiction.

3.2.4. Case $x \geq 4$. We take equation (4) modulo m^4 , so $r \equiv 1 \pmod{4}$ or $r \equiv 3 \pmod{4}$. We obtain $-\frac{1}{2}r(r-1)y \equiv z \pmod{m^2}$. Then we have

$$\frac{1}{2}r(r-1)y + z \geq m^2. \quad (39)$$

If $a^x \leq b^{\frac{y}{2}}$, then equation (4) and the condition $m \geq 24650r^2$ imply $z \log c < y \log b + \log 1.01$. We deduce that $z \leq \frac{1}{2}ry$. Using (29) and (39), we get $\frac{2m^2}{r^2} - 1.999 \leq y \leq 3040 \log(m^2 + 1)$. So one obtains $m < 200r \log r$. This contradicts the fact that $m > 24650r^2(\log r)^2$.

If $a^x > b^{\frac{y}{2}}$, then (30) and (31) give $y < \frac{7}{3}x$ and $z \leq rx$. Hence, from (39) we obtain

$$x \geq \frac{0.857}{r^2} m^2. \quad (40)$$

If $m = 2^s$ and as $m > 24650r^2$, then $s \geq 20$. So inequality (35) implies

$$x \leq 15.7 \left(\max \left\{ \log \left(\frac{2rx}{\log b} \right) + 0.4, 5 \right\} \right)^2 \log b \log c. \quad (41)$$

If $\log(\frac{2rx}{\log b}) \leq 4.6$, then we use (23) and (40) to obtain $m^2 \leq 60r \log b < 60r^2 \log m$, so $m < 30r \log r$. If $\log(\frac{2rx}{\log b}) > 4.6$, from (23) and $\sin \alpha > \frac{1}{2}\alpha$, we have $\log b < \frac{1}{2}r \log c < r \log m$ and $b > \frac{1}{2}c^{r/2} \frac{r}{m} > c^{\frac{r-2}{2}}$. So $\log b > \frac{r-2}{2} \log c > (r-2) \log m$. Thus inequality (41) becomes

$$x \leq 31.4 \left(\log \left(\frac{2.8x}{\log m} \right) + 0.4 \right)^2 r \log^2 m. \quad (42)$$

Hence, we combine inequalities (40) and (42) to obtain $m < 1000r^2(\log r)^2$. This is a contradiction.

If $m \neq 2^s$, then from inequality (36) and as $p \geq 3$ we have

$$x \leq 60 \left(\max \left\{ \log \left(\frac{2z}{\log b} \right) + 0.4, 6 \right\} \right)^2 \log b \log c. \quad (43)$$

If $\log(\frac{2rx}{\log b}) \leq 5.6$, then using (40) we have $m^2 \leq 271r \log b < 271r^2 \log m$. So we obtain $m < 91r \log r$. If $\log(\frac{2rx}{\log b}) > 5.6$, similarly from (40) and (43), we get $m < 1000r^2(\log r)^2$. This leads also to a contradiction. Therefore, the proof of Theorem 1.1 is complete.

4. Proofs of Theorem 1.2 and Theorem 1.3. Suppose that $r \equiv 5 \pmod{8}$ or $r \equiv 3, 19 \pmod{24}$, and $m \geq 2r$. Let $(x, y, z) \neq (2, 2, r)$ be a solution of equation (1). Then Lemma 2.10 implies that $x = 2$, $2 \mid y$, $y \geq 6$, and $2 \nmid z$. So equations (23) give $c^z = a^2 + b^y < c^r + c^{ry/2}(\sin(r\theta))^y < c^{ry/2}$. Thus we obtain $z < \frac{1}{2}ry$. Using equation (1), we have $\frac{1}{2}r(r-1)y + z \equiv r^2 \pmod{m^2}$. So if $r \equiv 5 \pmod{8}$ then we get $z \equiv 1 \pmod{4}$ and if $r \equiv 3, 19 \pmod{24}$ then we have $z \equiv 3 \pmod{4}$. We deduce that $\frac{1}{2}y(y-1)(\frac{1}{2}r(r-1))^2 - \frac{1}{2}z(z-1) \equiv 0 \pmod{2}$. Therefore, we consider equation (1) modulo $2m^2$ and we get $\frac{1}{2}r(r-1)y + z \equiv r^2 \pmod{2m^2}$. Since $y \geq 6$, then $\frac{1}{2}r(r-1)y + z - r^2 > 0$. So $\frac{1}{2}r(r-1)y + z - r^2 \geq 2m^2$. As $z < \frac{1}{2}ry$, we obtain

$$y \geq \frac{4}{r^2}m^2 + 2. \quad (44)$$

4.1. Proof of Theorem 1.2. Using Lemma 2.8 and Lemma 2.11, one can see that if $r \geq 18460$, then $c = m^2 + 1 \geq 1.37 \times 10^8$, so $b > c^{r/3}$. As $z > r$, using again Lemma 2.11 and equation (1), we obtain $b^{y/2} > c^{z/3}$ and $a > c^{z/3}$. Thus, by equations (1) and (2), we have

$$a > c^{z/3} = (a^2 + b^y)^{1/3} > b^{y/3} > c^{ry/9} = (a^2 + b^2)^{y/9} > a^{2y/9} > a. \quad (45)$$

This is impossible.

If $r < 18460$, then equation (4) and Lemma 2.8 imply that there exist $X_1, Y_1 \in \mathbb{N}$ such that $\gcd(X_1, Y_1) = 1$, $2 \mid X_1$, and

$$b^{y/2} = Y_1 \left| \sum_{k=0}^{(z-1)/2} \binom{z}{2k+1} X_1^{z-2k-1} (-Y_1^2)^k \right|, \quad (46)$$

where $c = m^2 + 1 = X_1^2 + Y_1^2$. If $Y_1 \neq 1$, then as $2 \nmid z$ and by (46), we have $Y_1 \mid b^{y/2}$.

Notice that $\gcd(m, b) = 1$. Thus there exists a prime $p \geq 11$ (see Lemma 2.9) such that $p \mid Y_1$. Hence, using equation (46) or Lemma 1 of [33], we have

$$\text{ord}_p b^{y/2} = \text{ord}_p Y_1 + \text{ord}_p z. \quad (47)$$

As $p \mid b$, then $\text{ord}_p b \geq 1$. Thus

$$\text{ord}_p z = \frac{y}{2} \text{ord}_p b - \text{ord}_p Y_1 \geq \frac{y}{2} - \text{ord}_p Y_1. \quad (48)$$

The identity $c = X_1^2 + Y_1^2$ implies $Y_1 \leq \sqrt{c-4}$. So $\text{ord}_p Y_1 \leq \log_p Y_1 < \log_p \sqrt{c-4}$. Using $\log_p z \geq \text{ord}_p z$ and inequality (48), we obtain $z > \frac{p^{y/2}}{\sqrt{c-4}}$. Also as $z < \frac{1}{2}ry$, we have

$$p^{y/2} < \frac{1}{2}ry\sqrt{c-4} < \frac{1}{2}rym. \quad (49)$$

We use (44) to get $m < \frac{1}{2}r\sqrt{y}$. Then from (49), $p \geq 11$ and $r < 18460$. Therefore, we have

$$11^{y/2} < \frac{1}{4}r^2y^{3/2} < 8.52 \times 10^8 y^{3/2}. \quad (50)$$

Notice that $f(y) = 11^{y/2} - 8.52 \times 10^8 y^{3/2}$ is an increasing function of y and $m \geq 2r$. Hence, inequality (44) gives $y \geq 20$. So $f(y) \geq f(20) \geq 11^{10} - 8.52 \times 10^8 \times 6^{3/2} > 0$. This contradicts inequality (50). Therefore, we consider $Y_1 = 1$. Thus from equations (1), (2), (46), and Lemma 2.8, we get

$$v_r + u_r y^{1/2} \sqrt{-1} = \pm(m + \sqrt{-1})^z. \quad (51)$$

Let $\alpha = m + \sqrt{-1}$, $\beta = m - \sqrt{-1}$. Then $\alpha + \beta$ and $\alpha\beta$ are nonzero, coprime integers, and $\frac{\alpha}{\beta}$ is a root of the equation $(m^2 + 1)X^2 - 2(m^2 - 1)X + m^2 + 1 = 0$. Since $\gcd(m^2 + 1, 2(m^2 - 1)) = 1$, then $\frac{\alpha}{\beta}$ is not a root of unity. So (α, β) is a Lucas pair. Therefore, we get

$$b^{y/2} = \pm \frac{\alpha^z - \beta^z}{\alpha - \beta}, \quad b^r = \pm \frac{\alpha^r - \beta^r}{\alpha - \beta}.$$

If $y \geq 6$, then $z > r$. So $b^{y/2}$ has no primitive divisor. If $z > 30$, by Lemma 2.15, it is impossible. If $z \neq 3, 5, 7$, using Lemma 2.15, one can see that it is also impossible. If $z = 3, 5, 7$, taking all solutions of Theorem 1 in [31], we verify again the impossibility. Then the equation has only the positive integer solution $(x, y, z) = (2, 2, r)$. So the proof of Theorem 1.2 is complete.

4.2. Proof of Theorem 1.3. As r is a prime and by Lemma 2.14, if p is a factor of b , then we have $r \mid (p-1)$ or $r \mid (p+1)$. Hence, for any prime p such that $p \mid b$, we get

$$p \geq 2r - 1. \quad (52)$$

If $Y_1 \neq 1$, then there exists a prime p such that $p \geq 2r - 1$ and $p \mid X_1$. From (48), (49), and (44), we have

$$(2r-1)^{y/2} < \frac{1}{2}ry\sqrt{c-4} < \frac{1}{2}r^2y^{3/2}. \quad (53)$$

As $r \geq 7$ and $y \geq 6$, we get a contradiction to (53). So we consider $Y_1 = 1$. A similar method to that used in the proof of Theorem 1.2 shows that it is impossible. This completes the proof of Theorem 1.3.

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